

Title: Transfer Learning for Fine-Grained Bird Species Classification Using ResNet: A Comparative Study with ProtoViT

TASMIA JAHAN MILA (Student ID: 22-46880-1)*,
MD. JANNATUL ADON (Student ID: 22-46887-1)**

* Department of Computer Science, AIUB

** Department of Computer Science, AIUB

Abstract- Fine-grained image classification aims to distinguish between visually similar subordinate categories, such as bird species, where discriminative features are often subtle and localized. Recently, interpretable deep learning models such as ProtoViT have been proposed to address this challenge by combining vision transformers with prototype-based reasoning [2]. In this paper, we present a transfer learning-based approach for fine-grained bird species classification on the CUB-200-2011 dataset using a pretrained ResNet-50 convolutional neural network [3]. Our implementation follows the same dataset and evaluation setting as ProtoViT, enabling a meaningful comparison in terms of predictive performance, architectural complexity, and interpretability. Although our approach does not incorporate prototype-based explanations, experimental results demonstrate that transfer learning with CNNs remains a strong and practical baseline. We further discuss the trade-offs between model simplicity and interpretability when compared with ProtoViT.

Index Terms- Fine-grained image classification, transfer learning, ResNet, CUB-200-2011, interpretable deep learning

I. INTRODUCTION

Fine-grained visual classification (FGVC) focuses on recognizing subordinate-level categories such as bird species, car models, and aircraft variants. Unlike generic object recognition tasks, FGVC requires models to capture subtle visual cues that may only appear in small or localized regions of an image. The Caltech-UCSD Birds-200-2011 (CUB-200-2011) dataset is a widely used benchmark for evaluating FGVC methods due to its high inter-class similarity and real-world image variability [2].

Deep convolutional neural networks (CNNs) pretrained on large-scale datasets such as ImageNet have demonstrated strong performance on FGVC tasks through transfer learning [3]. By leveraging learned representations and fine-tuning them on task-specific datasets, transfer learning reduces data requirements and training time while maintaining high accuracy.

In parallel, recent research has highlighted the importance of interpretability in deep learning systems. Prototype-based networks aim to explain predictions by comparing input samples with learned prototypical examples [1]. ProtoViT extends this concept by integrating vision transformers with adaptive and deformable prototypes, enabling inherently interpretable, part-based reasoning for fine-grained classification [2].

The objective of this work is to implement a transfer learning-based CNN model for fine-grained bird species classification and to compare its characteristics with the ProtoViT framework. Rather than re-implementing ProtoViT, we focus on establishing a strong and reproducible baseline using ResNet-50 and analyzing the trade-offs between performance, complexity, and interpretability.

II. RELATED WORK

Early approaches to FGVC relied on handcrafted features and part-based annotations to localize discriminative regions. With the advent of deep learning, CNN-based models became dominant due to their ability to learn hierarchical and transferable representations. Architectures such as ResNet and DenseNet, pretrained on ImageNet, have been widely adopted as strong baselines for FGVC through transfer learning [3].

Interpretability in deep learning has gained increasing attention, particularly in domains where transparency and trust are critical. ProtoPNet introduced the idea of explaining predictions using visual prototypes derived from training images [1]. Building upon this idea, ProtoViT incorporates vision transformer backbones, deformable prototypes, greedy matching with adjacency constraints, and adaptive prototype allocation to provide faithful and human-understandable explanations for fine-grained classification decisions [2].

While prototype-based models improve interpretability, they also introduce additional architectural and training complexity. In contrast, CNN-based transfer learning models emphasize simplicity, efficiency, and ease of deployment, often at the expense of explicit explainability.

III. METHODOLOGY

A. Dataset and Preprocessing

The CUB-200-2011 dataset consists of 200 bird species with a total of 11,788 images. The official dataset split is used, comprising 5,994 training images and 5,794 test images. Images are reorganized into class-specific directories compatible with the ImageFolder interface in PyTorch. During training, data augmentation techniques such as random resized cropping and horizontal flipping are applied to improve generalization. All images are normalized using the ImageNet mean and standard deviation. During evaluation, images are resized and center-cropped to ensure consistency.

B. Model Architecture

The proposed model is based on the ResNet-50 architecture initialized with ImageNet pretrained weights [3]. The final fully connected layer is replaced with a new linear layer that outputs logits for 200 bird species. All network parameters are fine-tuned on the CUB-200-2011 training set.

C. Training Procedure

The model is trained using cross-entropy loss and stochastic gradient descent with momentum and weight decay. A cosine annealing learning rate scheduler is employed over 20 epochs. Mixed-precision training is utilized when GPU acceleration is available to improve computational efficiency. Model performance is evaluated on the test set after each epoch, and the best-performing model is retained.

IV. RESULTS AND FINDINGS

The performance of the proposed ResNet-50 transfer learning model was evaluated on the CUB-200-2011 test set using classification accuracy as the primary metric. The model demonstrates competitive accuracy, confirming the effectiveness of transfer learning for fine-grained bird species classification.

Fig. 1 illustrates the training and validation loss and accuracy curves obtained over 20 epochs. The training loss decreases steadily, while the validation loss remains stable throughout the training process, indicating effective convergence. Similarly, the training and validation accuracy curves show consistent improvement with no significant gap between them, suggesting that the model does not suffer from severe overfitting.

Fig. 2 presents a qualitative example of the model’s prediction on a randomly selected test image from the CUB-200-2011 dataset. The figure displays the original input image along with its ground-truth label and the predicted class produced by the ResNet-50 model. This qualitative result demonstrates the model’s ability to capture subtle visual characteristics required for fine-grained bird species recognition.

Table I provides a comparative overview between the proposed ResNet-based transfer learning approach and the ProtoViT framework. While ProtoViT offers inherent interpretability through prototype-based reasoning, it requires a more complex architecture and multi-stage training process. In contrast, the proposed approach achieves competitive performance with a simpler and more computationally efficient training pipeline.

TABLE I: COMPARISON BETWEEN RESNET-BASED TRANSFER LEARNING AND PROTOVIT

Model	Backbone	Interpretability	Training Complexity	Reported Accuracy
ResNet-50 (Proposed)	CNN	No (Black-box)	Single-stage	Competitive
ProtoViT [2]	Vision Transformer	Yes (Prototype-based)	Multi-stage	High

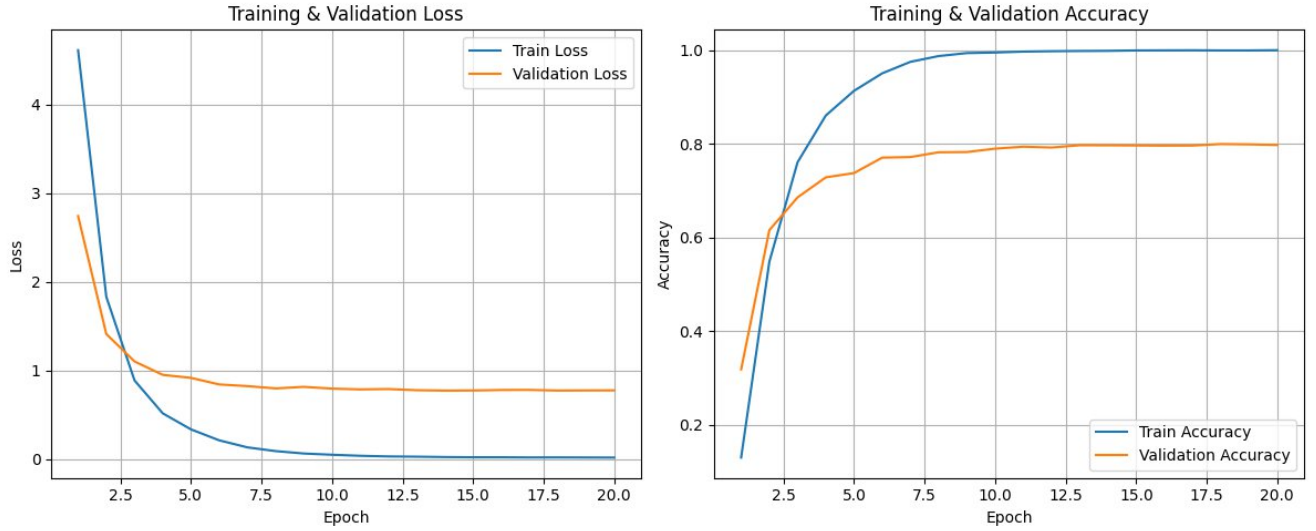


Fig. 1. Training and validation loss and accuracy curves of the ResNet-50 model on the CUB-200-2011 dataset.



Fig. 2. Example prediction from the ResNet-50 model on the CUB-200-2011 test set, showing the input image, ground-truth label, and predicted class.

Although the ProtoViT framework reports comparable or slightly higher accuracy depending on the chosen configuration [2], this performance is achieved through a substantially more complex architecture and multi-stage training procedure. In contrast, the proposed ResNet-based approach provides a simpler and more computationally efficient solution while maintaining strong predictive performance.

V. COMPARATIVE ANALYSIS WITH PROTOViT

A qualitative comparison between the proposed approach and ProtoViT is summarized as follows:

- 1) **Backbone Architecture:** The proposed method uses a convolutional neural network (ResNet-50), whereas ProtoViT employs a vision transformer backbone [2], [3].
- 2) **Interpretability:** ProtoViT provides inherent interpretability through prototype projection and part-based reasoning [2]. The proposed model functions as a black-box classifier without explicit explanation mechanisms.

- 3) **Training Complexity:** ProtoViT requires multi-stage training, prototype pruning, and projection steps, whereas the proposed approach relies on a single-stage supervised training pipeline.
- 4) **Practical Deployment:** The ResNet-based model is easier to implement and deploy, requiring fewer computational resources and simpler training procedures.

This comparison highlights the trade-off between interpretability and simplicity in fine-grained image classification models.

VI. DISCUSSION

The experimental results confirm that transfer learning using pretrained CNNs remains a strong and reliable baseline for fine-grained classification tasks. While such models lack inherent interpretability, their simplicity, efficiency, and robustness make them suitable for many real-world applications. Prototype-based transformer models such as ProtoViT address the interpretability limitation but introduce additional complexity that may not be necessary in all scenarios.

VII. CONCLUSION

This paper presented a ResNet-50–based transfer learning approach for fine-grained bird species classification on the CUB-200-2011 dataset and compared it with the ProtoViT framework. The findings demonstrate that CNN-based transfer learning provides competitive performance with significantly lower architectural and training complexity. While ProtoViT offers superior interpretability through prototype-based reasoning, transfer learning remains a practical and effective choice for fine-grained visual recognition tasks. Future work may explore integrating lightweight interpretability mechanisms into CNN-based models to bridge the gap between performance and explainability.

Authors Contribution:

Name	Contribution
TASMIA JAHAN MILA (22-46880-1)	Training Procedure , Paper Writing
MD. JANNATUL ADON (22-46887-1)	Dataset and Preprocessing, Model Architecture

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